

```

        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 828
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 837
    :constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
    }
    }
    :location 835
    }
    }
    :location 223
    }
    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname s_name
:ressortgroupref 2
:resorigtbl 17484
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 67
}
:resno 2
:resname numwait
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 410
:sortop 413
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowfuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
}

```

```

:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier_secure
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
        }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 17484
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname l1
        :colnames <>
        }
    :eref
      {ALIAS
        :aliasname l1
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_qua
          ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
          tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
          l_shipmode" "l_comment")
        }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 17488
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 2
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
          _orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
        }
    :rtekind 0
    :relid 17337
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 3
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
        }
    :rtekind 0
    :relid 17285
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 4
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname *GROUP*
        :colnames ("s_name")
        }
    :rtekind 9
    :groupexprs (
      {VAR
        :varno 8
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 8
        :varattnosyn 2
        :location -1
        }
      )
    :lateral false
    :inFromCl false
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname l2
        :colnames <>
        }
    :eref
      {ALIAS
        :aliasname l2
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_qua
          ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
          tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
          l_shipmode" "l_comment")
        }
    :rtekind 0
    :relid 17349
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 5
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname l3
        :colnames <>
        }
    :eref
      {ALIAS

```

```

:aliasname l3
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 17308
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17285
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17291
:inh false
:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 10
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17337
:inh false
:relkind r
:rellockmode 1
:perminfoindex 11
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 17318
:inh false
:relkind r
:rellockmode 1
:perminfoindex 12
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)

```

```

)
{RTEPERMISSIONINFO
:relid 17484
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17488
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 19 20)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17285
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17308
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17285
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17291
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17318
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
}
)
}

:jointree
{FROMEXPR
:fromlist (
:JOINEXPR
:jointype 5
:isNatural false
:larg
:JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
:JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 8
}
)
}
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 8
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
}
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 9
}
{RANGETBLREF
:rtindex 10
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 1
:location -1
}
{VAR
:varno 8
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 4
:location -1
}
)
}
:location -1
}
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 3
:location -1
}
{VAR
:varno 10
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 1
:location -1
}
)
}
:location -1
}
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 10
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
}
:location -1
}
}
:alias <>
:rtindex 0

```

supplier.s_acctbal > 0

n.n_nationkey =
supplier.s_nationkey
AND
n.n_regionkey =
r.r_regionkey
AND r.r_name =
'EUROPE'

```

}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 11
}
)
:quals <>
}
}
:rarg
{RANGETBLREF
:rtindex 12
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 12
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 12
:varattnosyn 1
:location -1
}
{VAR
:varno 11
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 1
:location -1
}
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 12
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 12
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0
2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 4
0 0 0 0 50 45 72 73 71 72 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 13
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 13
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 13
:varattnosyn 1
:location -1
}
{VAR
:varno 11
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 2
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 96
:opfuncid 65

```

```

:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 13
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 13
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 11
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 11
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 13
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 13
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 8
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 8
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 11
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 11
    :varattnosyn 3
    :location -1
  }
)
:location 200
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 227
  }
  {VAR
    :varno 11
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 11

```

```

ps.ps_partkey =
lineitem.l_partkey
ps.ps_suppkey =
lineitem.l_suppkey
ps.ps_availqty > 0

```

```

s_suppkey =
l1.l_suppkey
and o_orderkey =
l1.l_orderkey
and o_orderstatus
= 'F'
and
l1.l_receiptdate >
l1.l_commitdate
and s_nationkey =
n_nationkey
and n_name =
'ARGENTINA'

```

main query quals

```

        :varattnosyn 1
        :location -1
    }
}
:location 238
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 3
        :varattno 3
        :vartype 1042
        :vartypmod 5
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location 266
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location 282
        :constvalue 5 [ 20 0 0 0 70 ]
    }
)
:location 280
}
{OPEXPR
:opno 1097
:opfuncid 1009
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 11
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 13
        :location -1
    }
    {VAR
        :varno 11
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 12
        :location -1
    }
)
:location 315
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 8
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 8
        :varattnosyn 4
        :location -1
    }
    {VAR
        :varno 4
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location 804
    }
)
:location 802
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 828
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location 837
        :constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
    }
)
:location 835
}

```

```

    )
  }
  :rarg
  {RANGETBLREF
   :rtindex 6
  }
  :usingClause <>
  :join_using_alias <>
  :quals (
    {OPEXPR
     :opno 96
     :opfuncid 65
     :opresulttype 16
     :opretset false
     :opcollid 0
     :inputcollid 0
     :args (
       {VAR
        :varno 6
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 6
        :varattnosyn 1
        :location 434
       }
       {VAR
        :varno 11
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 1
        :location -1
       }
     )
     :location 448
   }
  {OPEXPR
   :opno 518
   :opfuncid 144
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 6
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 6
      :varattnosyn 3
      :location 486
     }
     {VAR
      :varno 11
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 11
      :varattnosyn 3
      :location -1
     }
   )
   :location 499
  }
  :alias <>
  :rtindex 0
  }
  :rarg
  {RANGETBLREF
   :rtindex 7
  }
  :usingClause <>
  :join_using_alias <>
  :quals (
    {OPEXPR
     :opno 96
     :opfuncid 65
     :opresulttype 16
     :opretset false
     :opcollid 0
     :inputcollid 0
     :args (
       {VAR
        :varno 7
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 7
        :varattnosyn 1
        :location 630
       }
       {VAR
        :varno 11
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 11
        :varattnosyn 1
        :location -1
       }
     )
     :location 644
   }
  {OPEXPR
   :opno 518
   :opfuncid 144
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 7
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
    }
  )

```

```

:varnosyn 7
:varattnosyn 3
:location 682
}
{VAR
:varno 11
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 3
:location -1
}
)
:location 695
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 13
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 13
:location 733
}
{VAR
:varno 7
:varattno 12
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 12
:location 752
}
)
:location 750
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 8
:varattno 2
:varatype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 2
:location -1
}
:resno 1
:resname s_name
:ressortgroupref 2
:resorigtbl 17484
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 67
}
:resno 2
:resname numwait
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 410
:sortop 413
:reverse_sort true
}
)

```

```
WARNING: is_parallel_safe: OUTPUT qual = {{RESTRICTINFO: clause {OEXPR : opno 1097 : opfuncid 1089 : opresulttype 16 : opreset false : opcollid 0 : inputcollid 0 : args ({VAR : varno 11 : varattno 13 : vartype 1082 : vartypmod -1 : varcollid 0 : varnullingrels (b) : varlevelsup 0 : varreturningtype 0 : varnosyn 11 : varattnosyn 13 : location -1} {VAR : varno 11 : varattno 12 : vartype 1082 : vartypmod -1 : varcollid 0 : v
```

[illegible]

[illegible]

```
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 8 :varattno 2 :vartype 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 8 :varattnosyn 2 :location -1) {VAR :varno 8 :varattno 4 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 8 :varattnosyn 4 :location -1} {VAR :varno 8 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 8 :varattnosyn 1 :location -1}) (VAR :varno 13 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 13 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varsynsyn 4 :varattnosyn 1 :location
```


[illegible]

[illegible]

```
WARNING: is_parallel_safe: OUTPUT qual = (VAR :varno 8 :varattno 8 :varpartno 1042 :vartypemod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattnosyn 2 :location -1) (VAR :varno 8 :varattno 4 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattnosyn 4 :location -1) (VAR :varno 9 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattnosyn 1 :location -1) (VAR :varno 11 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattnosyn 1 :location -1) (VAR :varno 11 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattnosyn 3 :location -1) (VAR :varno 11 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattnosyn 3 :location -1)
```

[illegible]

[illegible]

[illegible]

[illegible]

```
Sort (cost=175745.04..175745.04 rows=1 width=34) (actual time=454.338..461.070 rows=0.00 loops=1)
  Output: supplier.s_name, (count(*))
  Sort Key: (count(*)) DESC, supplier.s_name
  Sort Method: quicksort Memory: 25kB
  Buffers: shared hit=5514 read=41694, temp read=5791 written=5864
  -> GroupAggregate (cost=175745.05..175745.03 rows=1 width=34) (actual time=454.317..461.048 rows=0.00 loops=1)
    Output: supplier.s_name, count(*)
    Group Key: supplier.s_name
    Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
    -> Sort (cost=175745.01..175745.01 rows=1 width=26) (actual time=454.315..461.047 rows=0.00 loops=1)
      Output: supplier.s_name
      Sort Key: supplier.s_name
      Sort Method: quicksort Memory: 25kB
      Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
      -> Gather (cost=131615.19..175745.00 rows=1 width=26) (actual time=454.309..461.040 rows=0.00 loops=1)
        Output: supplier.s_name
        Workers Planned: 2
        Workers Launched: 2
        Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
        -> Parallel Hash Right Anti Join (cost=130615.19..174744.90 rows=1 width=26) (actual time=451.290..451.299 rows=0.00 loops=3)
          Output: supplier.s_name
          Hash Cond: (l3.l_orderkey = lineitem.l_orderkey)
          Join Filter: (l3.l_suppkey <> lineitem.l_suppkey)
          Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
          Worker 0:  actual time=450.171..450.179 rows=0.00 loops=1
            Buffers: shared hit=1814 read=13687, temp read=1923 written=1928
          Worker 1:  actual time=450.281..450.289 rows=0.00 loops=1
            Buffers: shared hit=1834 read=13986, temp read=1949 written=1956
          -> Parallel Seq Scan on public.lineitem l3 (cost=0.00..43192.09 rows=250029 width=8) (never executed)
            Output: l3.l_orderkey, l3.l_partkey, l3.l_suppkey, l3.l_linenum, l3.l_quantity, l3.l_extendedprice, l3.l_discount, l3.l_tax, l3.l_returnflag, l3.l_linestatus, l3.l_shipdate, l3.l_commitdate, l3.l_receiptdate, l3.l_shipinstruct, l3.l_shipmode, l3.l_comment
            Filter: (l3.l_receiptdate > l3.l_commitdate)
          -> Parallel Hash (cost=130615.18..130615.18 rows=1 width=34) (actual time=451.275..451.284 rows=0.00 loops=3)
            Output: supplier.s_name, lineitem.l_orderkey, lineitem.l_suppkey
            Buckets: 1024 Batches: 1 Memory Usage: 0kB
            Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
            Worker 0:  actual time=450.164..450.172 rows=0.00 loops=1
              Buffers: shared hit=1814 read=13687, temp read=1923 written=1928
            Worker 1:  actual time=450.190..450.197 rows=0.00 loops=1
              Buffers: shared hit=1834 read=13986, temp read=1949 written=1956
            -> Parallel Hash Semi Join (cost=117101.60..130615.18 rows=1 width=34) (actual time=451.149..451.158 rows=0.00 loops=3)
              Output: supplier.s_name, lineitem.l_orderkey, lineitem.l_suppkey
```

```
Hash Cond: (o.o_orderkey = l2.l_orderkey)
Join Filter: (l2.l_suppkey <> lineitem.l_suppkey)
Buffers: shared hit=5508 read=41694, temp read=5791 written=5864
Worker 0: actual time=449.978..449.985 rows=0.00 loops=1
  Buffers: shared hit=1814 read=13687, temp read=1923 written=1928
Worker 1: actual time=450.175..450.182 rows=0.00 loops=1
  Buffers: shared hit=1834 read=13986, temp read=1949 written=1956
-> Parallel Hash Semi Join (cost=63477.64..74058.20 rows=2 width=42) (actual time=162.627..162.635 rows=0.00 loops=3)
  Output: supplier.s_name, lineitem.l_orderkey, lineitem.l_suppkey, o.o_orderkey, orders.o_orderkey
  Hash Cond: ((lineitem.l_partkey = ps.ps_partkey) AND (lineitem.l_suppkey = ps.ps_suppkey))
  Buffers: shared hit=3883 read=9583
  Worker 0: actual time=163.461..163.467 rows=0.00 loops=1
    Buffers: shared hit=1288 read=3073
  Worker 1: actual time=162.207..162.213 rows=0.00 loops=1
    Buffers: shared hit=1302 read=3238
-> Parallel Hash Join (cost=55455.79..66036.31 rows=2 width=50) (actual time=96.465..96.472 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, o.o_orderkey, orders.o_orderkey
  Buffers: shared hit=2285 read=5829
  Worker 0: actual time=96.474..96.480 rows=0.00 loops=1
    Buffers: shared hit=763 read=1923
  Worker 1: actual time=96.462..96.467 rows=0.00 loops=1
    Buffers: shared hit=763 read=1946
-> Parallel Seq Scan on public.orders (cost=0.00..10239.75 rows=90869 width=4) (never executed)
  Output: orders.o_orderkey, orders.o_custkey, orders.o_orderstatus, orders.o_totalprice, orders.o_orderdate, orders.o_orderpriority, orders.o_cler
k, orders.o_shippriority, orders.o_comment
  Filter: (orders.o_orderstatus = 'F')::bpchar)
-> Parallel Hash (cost=55455.75..55455.75 rows=3 width=46) (actual time=96.437..96.443 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, o.o_orderkey
  Buckets: 1024 Batches: 1 Memory Usage: 0kB
  Buffers: shared hit=2285 read=5829
  Worker 0: actual time=96.432..96.437 rows=0.00 loops=1
    Buffers: shared hit=763 read=1923
  Worker 1: actual time=96.448..96.453 rows=0.00 loops=1
    Buffers: shared hit=763 read=1946
-> Parallel Hash Semi Join (cost=11325.17..55455.75 rows=3 width=46) (actual time=95.790..95.796 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, o.o_orderkey
  Hash Cond: (lineitem.l_orderkey = o.o_orderkey)
  Buffers: shared hit=2285 read=5829
  Worker 0: actual time=96.422..96.427 rows=0.00 loops=1
    Buffers: shared hit=763 read=1923
  Worker 1: actual time=95.544..95.548 rows=0.00 loops=1
    Buffers: shared hit=763 read=1946
-> Hash Join (cost=148.62..44279.15 rows=8 width=42) (actual time=1.075..1.079 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey
  Hash Cond: (lineitem.l_suppkey = supplier.s_suppkey)
  Buffers: shared hit=212 read=6
  Worker 0: actual time=1.338..1.342 rows=0.00 loops=1
    Buffers: shared hit=70 read=3
  Worker 1: actual time=1.009..1.012 rows=0.00 loops=1
    Buffers: shared hit=70 read=3
-> Parallel Seq Scan on public.lineitem (cost=0.00..43192.09 rows=250029 width=12) (actual time=0.166..0.166 rows=1.00 loops=3)
  Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_ext
endedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_returnflag, lineitem.l_linestatus, lineitem.l_shipdate, lineitem.l_commitdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shi
pmode, lineitem.l_comment
  Filter: (lineitem.l_receiptdate > lineitem.l_commitdate)
  Buffers: shared hit=2 read=6
  Worker 0: actual time=0.429..0.429 rows=1.00 loops=1
    Buffers: shared read=3
  Worker 1: actual time=0.055..0.056 rows=1.00 loops=1
    Buffers: shared read=3
-> Hash (cost=148.61..148.61 rows=1 width=30) (actual time=0.888..0.891 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey
  Buckets: 1024 Batches: 1 Memory Usage: 0kB
  Buffers: shared hit=210
  Worker 0: actual time=0.885..0.888 rows=0.00 loops=1
    Buffers: shared hit=70
  Worker 1: actual time=0.935..0.938 rows=0.00 loops=1
    Buffers: shared hit=70
-> Nested Loop (cost=24.50..148.61 rows=1 width=30) (actual time=0.888..0.890 rows=0.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey
  Join Filter: (nation.n_nationkey = n.n_nationkey)
  Rows Removed by Join Filter: 510
  Buffers: shared hit=210
  Worker 0: actual time=0.885..0.888 rows=0.00 loops=1
    Buffers: shared hit=70
  Worker 1: actual time=0.935..0.938 rows=0.00 loops=1
    Buffers: shared hit=70
-> Seq Scan on public.nation (cost=0.00..12.12 rows=1 width=4) (actual time=0.050..0.053 rows=1.00 loops=3)
  Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
  Filter: (nation.n_name = 'ARGENTINA')::bpchar)
  Rows Removed by Filter: 24
  Buffers: shared hit=3
  Worker 0: actual time=0.053..0.056 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.074..0.076 rows=1.00 loops=1
    Buffers: shared hit=1
-> Hash Semi Join (cost=24.50..136.28 rows=16 width=38) (actual time=0.097..0.811 rows=510.00 loops=3)
  Output: supplier.s_name, supplier.s_nationkey, supplier.s_suppkey, n.n_nationkey
  Hash Cond: (supplier.s_nationkey = n.n_nationkey)
  Buffers: shared hit=207
  Worker 0: actual time=0.110..0.806 rows=510.00 loops=1
    Buffers: shared hit=69
  Worker 1: actual time=0.126..0.835 rows=510.00 loops=1
    Buffers: shared hit=69
-> Seq Scan on public.supplier (cost=0.00..104.50 rows=2708 width=34) (actual time=0.031..0.560 rows=2714.00 loop
s=3)
  Output: supplier.s_suppkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supp
lier.s_acctbal, supplier.s_comment
  Filter: (supplier.s_acctbal > '0')::numeric)
  Rows Removed by Filter: 286
  Buffers: shared hit=201
  Worker 0: actual time=0.037..0.548 rows=2714.00 loops=1
    Buffers: shared hit=67
  Worker 1: actual time=0.049..0.571 rows=2714.00 loops=1
    Buffers: shared hit=67
-> Hash (cost=24.48..24.48 rows=1 width=4) (actual time=0.055..0.057 rows=5.00 loops=3)
  Output: n.n_nationkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=6
  Worker 0: actual time=0.063..0.064 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1: actual time=0.067..0.068 rows=5.00 loops=1
    Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.048..0.053 rows=5.00 loops=3)
  Output: n.n_nationkey
  Hash Cond: (n.n_regionkey = r.r_regionkey)
  Buffers: shared hit=6
  Worker 0: actual time=0.056..0.060 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1: actual time=0.060..0.064 rows=5.00 loops=1
    Buffers: shared hit=2
-> Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.002..0.003 rows=25.
00 loops=3)
  Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
  Buffers: shared hit=3
  Worker 0: actual time=0.002..0.003 rows=25.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.002..0.003 rows=25.00 loops=1
    Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.037..0.037 rows=1.00 loops=3)
  Output: r.r_regionkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=3
  Worker 0: actual time=0.044..0.045 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.049..0.050 rows=1.00 loops=1
    Buffers: shared hit=1
-> Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.030..0.030 rows
=1.00 loops=3)
  Output: r.r_regionkey
  Filter: (r.r_name = 'EUROPE')::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=3
  Worker 0: actual time=0.036..0.037 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.044..0.044 rows=1.00 loops=1
    Buffers: shared hit=1
```

```
-> Parallel Hash (cost=10239.75..10239.75 rows=74944 width=4) (actual time=94.384..94.385 rows=60149.33 loops=3)
  Output: o.o_orderkey
  Buckets: 262144 Batches: 1 Memory Usage: 9184kB
  Buffers: shared hit=2073 read=5823
  Worker 0: actual time=94.512..94.513 rows=59909.00 loops=1
    Buffers: shared hit=693 read=1920
  Worker 1: actual time=94.126..94.126 rows=59752.00 loops=1
    Buffers: shared hit=693 read=1943
-> Parallel Seq Scan on public.orders o (cost=0.00..10239.75 rows=74944 width=4) (actual time=0.774..84.047 rows=60149.33 loops=3)
  Output: o.o_orderkey
  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
  Rows Removed by Filter: 89851
  Buffers: shared hit=2073 read=5823
  Worker 0: actual time=0.900..84.084 rows=59909.00 loops=1
    Buffers: shared hit=693 read=1920
  Worker 1: actual time=0.893..84.261 rows=59752.00 loops=1
    Buffers: shared hit=693 read=1943
-> Parallel Hash (cost=6522.00..6522.00 rows=99990 width=8) (actual time=65.593..65.593 rows=80000.00 loops=3)
  Output: ps.ps_partkey, ps.ps_suppkey
  Buckets: 262144 Batches: 1 Memory Usage: 11488kB
  Buffers: shared hit=1598 read=3674
  Worker 0: actual time=65.728..65.728 rows=75989.00 loops=1
    Buffers: shared hit=525 read=1150
  Worker 1: actual time=65.329..65.329 rows=83504.00 loops=1
    Buffers: shared hit=539 read=1292
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..6522.00 rows=99990 width=8) (actual time=0.693..53.037 rows=80000.00 loops=3)
  Output: ps.ps_partkey, ps.ps_suppkey
  Filter: (ps.ps_availqty > 0)
  Buffers: shared hit=1598 read=3674
  Worker 0: actual time=0.702..53.736 rows=75989.00 loops=1
    Buffers: shared hit=525 read=1150
  Worker 1: actual time=1.007..52.541 rows=83504.00 loops=1
    Buffers: shared hit=539 read=1292
-> Parallel Hash (cost=41316.87..41316.87 rows=750087 width=8) (actual time=236.209..236.209 rows=600031.00 loops=3)
  Output: l2.l_orderkey, l2.l_suppkey
  Buckets: 262144 Batches: 16 Memory Usage: 6528kB
  Buffers: shared hit=1625 read=32191, temp written=5864
  Worker 0: actual time=235.230..235.230 rows=593202.00 loops=1
    Buffers: shared hit=526 read=10614, temp written=1928
  Worker 1: actual time=235.256..235.257 rows=599968.00 loops=1
    Buffers: shared hit=532 read=10748, temp written=1956
-> Parallel Seq Scan on public.lineitem l2 (cost=0.00..41316.87 rows=750087 width=8) (actual time=0.502..135.956 rows=600031.00 loops=3)
  Output: l2.l_orderkey, l2.l_suppkey
  Buffers: shared hit=1625 read=32191
  Worker 0: actual time=0.869..136.897 rows=593202.00 loops=1
    Buffers: shared hit=526 read=10614
  Worker 1: actual time=0.611..135.474 rows=599968.00 loops=1
    Buffers: shared hit=532 read=10748
```

Planning:
 Buffers: shared hit=271
Planning Time: 29.005 ms
Execution Time: 461.340 ms
(234 rows)

```
=====
BENCHMARK SUMMARY
=====
Runs          : 5
Average Planning : 29.635 ms
Average Execution : 495.383 ms

Generated Files:
  run5_explain.txt
  summary.txt
adityadave@Adityas-MacBook-Air-3 Q2&Q21 %
```